

PRACTICE SHEET

- The decimal equivalent of $(101011)_2$ is:
 (a) $(43)_{10}$ (b) $(59)_{10}$
 (c) $(47)_{10}$ (d) None of these
- $100101 + 101 + 1101 + 100$ is equal to:
 (a) 1101001 (b) 100000
 (c) 111011 (d) None of these
 - What is the decimal number representation of the binary number $(11101.001)_2$?
 (a) 30.125 (b) 29.025
 (c) 29.125 (d) 28.025
 - Binary fraction 0.1011 in decimal system is:
 (a) 0.6875 (b) 0.8675
 (c) 0.7685 (d) None of these
 - The decimal number 0.77 in binary system is:
 (a) $(0.1100101)_2$ (b) $(0.11000101)_2$
 (c) $(0.10100101)_2$ (d) None of these
 - The binary number 10110100001 in decimal system is:
 (a) 441 (b) 1441
 (c) 1241 (d) 241
 - The complement of $(1011100)_2$ is:
 (a) $(1010100)_2$ (b) $(0111100)_2$
 (c) $(0100011)_2$ (d) $(1000011)_2$
 - 2's complement of $(1011001100)_2$ is:
 (a) $(0100110100)_2$ (b) $(0100110010)_2$
 (c) $(0100110101)_2$ (d) $(0100110111)_2$
 - What is the decimal equivalent of $(101.101)_2$?
 (a) $(5.225)_{10}$ (b) $(5.525)_{10}$
 (c) $(5.625)_{10}$ (d) $(5.65)_{10}$
 - If the sum of the binary numbers $(11011)_2$, $(10110110)_2$ and $(10011x0y)_2$ is the binary number $(101101101)_2$, then the values of x and y respectively, are:
 (a) 1 and 1 (b) 1 and 0
 (c) 0 and 1 (d) 0 and 0
 - What is the value of X, if $(1010)_2 \times (111)_2 = (X)_{10}$?
 (a) 60 (b) 70
 (c) 75 (d) 80
 - What is $(1111)_2 + (1001)_2 - (1010)_2$ equal to?
 (a) $(111)_2$ (b) $(1100)_2$
 (c) $(1110)_2$ (d) $(1010)_2$
 - The difference of two numbers $(1100110011)_2$ and $(1101001011)_2$ in binary system is:
 (a) $(100000)_2$ (b) $(101010)_2$
 (c) $(11000)_2$ (d) $(10111)_2$
 - The maximum three digit integer in the decimal system will be represented in the binary system by which one of the following?
 (a) $(1111110001)_2$ (b) $(111111110)_2$
 (c) $(1111100111)_2$ (d) $(1111000111)_2$
 - What is $(1111)_2 + (1001)_2 - (1010)_2$ equal to?
 (a) $(111)_2$ (b) $(1100)_2$
 (c) $(1110)_2$ (d) $(1010)_2$
 - What is the binary number equivalent of the decimal number 32.25?
 (a) 100010.10 (b) 100000.10
 (c) 100010.01 (d) 100000.01
 - What is the equivalent binary number of the decimal number 13.625?
 (a) 1101.111 (b) 1111.101
 (c) 1101.101 (d) 1111.111
 - The number 0.0011 in binary system represents?
 (a) Rational number $3/8$ in decimal system
 (b) Rational number $1/8$ in decimal system
 (c) Rational number $3/16$ in decimal system
 (d) Rational number $5/16$ in decimal system
 - When is the value of:

$$\frac{(0.101)_2^{(11)} + (0.011)_2^{(11)}}{(0.101)_2^{(10)} - (0.101)_2^{(01)} (0.011)_2^{(01)} + (0.011)_2^{(10)}}$$
 (a) $(0.001)_2$ (b) $(0.01)_2$
 (c) $(0.1)_2$ (d) $(1)_2$
 - If $x = (1101)_2$ and $y = (110)_2$, then what is the value of $x^2 - y^2$?
 (a) $(1000101)_2$ (b) $(10000101)_2$
 (c) $(10001101)_2$ (d) $(10010101)_2$

ANSWER KEYS

1.	a	2.	c	3.	c	4.	a	5.	b	6.	b	7.	c	8.	a	9.	c	10.	b
11.	b	12.	c	13.	c	14.	c	15.	c	16.	d	17.	c	18.	c	19.	d	20.	b

Solutions

Sol. 1. (a)

$$(101011)_2 = (?)_{10}$$

$$= [1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0]_{10}$$

$$= [32 + 0 + 8 + 0 + 2 + 1]_{10} = (43)_{10}$$

Sol. 2. (c)

$$\begin{array}{r} 100101 \\ + 101 \\ + 1101 \\ + 100 \\ \hline \end{array}$$

$$= 1110011$$

Sol. 3. (c)

$$(11101.001)_2 = (?)_{10}$$

$$= [1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 + 0 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3}]_{10}$$

$$= [16 + 8 + 4 + 1 + 0.125]_{10} = (29.125)_{10}$$

Sol. 4. (a)

$$= [1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3} + 1 \times 2^{-4}]_{10}$$

$$= 11/16 = 0.6875$$

Sol. 5. (b)

2	0.77	
2	1.54	1
2	1.08	1
2	0.16	0
2	0.32	0
2	0.64	0
	1.28	1
	0.56	0
	1.12	1

$$(0.77)_2 = (0.11000101)_{10}$$

Sol. 6. (b)

$$[1 \times 2^{10} + 0 \times 2^9 + 1 \times 2^8 + 1 \times 2^7 + 0 \times 2^6 + 1 \times 2^5 + 0 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0]_{10}$$

$$= [1024 + 256 + 128 + 32 + 1] = 1441$$

Sol. 7. (c)

complement of 1011100 is 0100011

Sol. 8. (a)

1st complement of 1011001100 is 0100110011

2nd complement = 1st complement + 1

$$= 0100110011 + 0100110100$$

Sol. 9. (c)

$$(101.101)_2 = (?)_{10}$$

$$= [1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 + 1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3}]_{10}$$

$$= [4 + 1 + 0.625]_{10} = (5.625)_{10}$$

Sol. 10. (b)

$$\begin{array}{r} 11011 \\ + 10110110 \\ + 10011x0y \\ \hline 101101101 \end{array}$$

$$y = 0, x = 1$$

Sol. 11. (b)

$$1010$$

$$\times 111$$

$$1010$$

$$1010 \times$$

$$1010 \times$$

$$1000110$$

$$(1000110)_2 = (70)_{10}$$

Sol. 12. (c)

$$\begin{array}{r} 1111 \\ + 1001 \\ \hline \end{array}$$

$$11000$$

$$- 1010$$

$$= 1110$$

Sol. 13. (c)

$$\begin{array}{r} 1101001011 \\ - 1100110011 \\ \hline \end{array}$$

$$= 11000$$

Sol. 14. (c)

2	999	1
2	499	1
2	249	1
2	124	0
2	62	0
2	31	1
2	15	1
2	7	1
2	3	1
	1	1

$$(999)_2 = (1111100111)_2$$

Sol. 15. (c)

$$\begin{array}{r} 1111 \\ + 1001 \\ \hline \end{array}$$

$$= 11000$$

$$- 1010$$

$$= 1110$$

Sol. 16. (d)

For integer part of 57.375 i.e., $(57)_{10}$

2	32	0
2	16	0
2	8	0
2	4	0
2	2	0
	1	1

$$\therefore (32)_{10} = (100000)_2$$

For after decimal part of 32.25 i.e., $(0.25)_{10}$

Now,

$$0.25 \times 2 = 0.50$$

$$0.50 \times 2 = 1.00$$

$$(0.25)_{10} = (10)_{2}$$

Binary

$$0$$

$$1$$

$$\therefore (32.25)_{10} = (100000.01)_2$$

Sol. 17. (c)

For integer part of 13.625 i.e., $(13)_{10}$

2	13	1
2	6	0
2	3	1
2	1	1

$$\therefore (13)_{10} = (1101)_2$$

For after decimal part of 32.625 i.e., $(0.625)_{10}$

Now,

Binary

$$0.625 \times 2 = 1.25 \quad 1$$

$$0.25 \times 2 = 0.50 \quad 0$$

$$0.50 \times 2 = 1.00 \quad 1$$

$$(0.625)_{10} = (0.101)_2$$

$$\therefore (13.625)_{10} = (1101.101)_2$$

Sol. 18. (c)

$$(0.0011)_2 = 2^{-1} \times 0 + 2^{-2} \times 0 + 2^{-3} \times 1 + 2^{-4} \times 1 =$$

$$\frac{1}{8} + \frac{1}{16} = \frac{3}{16}$$

Sol. 19. (d)

$\therefore$

$$\begin{aligned} & \frac{(0.101)_{10}^{(10)}_2 + (0.011)_{10}^{(11)}_2}{(0.101)_{10}^{(10)}_2 - (0.101)_{10}^{(01)}_2 (0.011)_{10}^{(01)}_2 + (0.011)_{10}^{(10)}_2} \\ &= \frac{\left(\frac{5}{8}\right)^3 + \left(\frac{3}{8}\right)^3}{\left(\frac{5}{8}\right)^2 - \left(\frac{5}{8}\right)\left(\frac{3}{8}\right) + \left(\frac{3}{8}\right)^2} \end{aligned}$$

$$= \frac{5}{8} + \frac{3}{8} = \frac{8}{8} = 1 = (1)_2$$

Sol. 20. (b)

$$\text{Given, } x = (1101)_2 = 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$$

$$= 8 + 4 + 1 = 13$$

$$\text{and } y = (110)_2 = 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$$

$$= 4 + 2 = 6$$

$$\therefore x^2 - y^2 = (13)^2 - (6)^2 = 169 - 36 = 133$$

Now,

2	133	
2	66	1
2	33	0
2	16	1
2	8	0
2	4	0
2	2	0
	1	0

$$\therefore 133 = (1000101)_2$$

NDA PYQ

1. The number 10101111 in binary system is represented in decimal system by which one of the following numbers?
(a) 157 (b) 175
(c) 571 (d) 751
[NDA 2011 (1)]
2. In the binary system of numbers let $a = 00111$ and $b = 01110$, then in decimal system what is b/a equal to?
(a) 1 (b) 2
(c) 4 (d) 5
[NDA 2011(1)]
3. In binary system decimal number 0.3 can be expressed as:
(a) $(0.01001\dots)_2$ (b) $(0.10110\dots)_2$
(c) $(0.11001\dots)_2$ (d) $(0.10101\dots)_2$
[NDA 2011(2)]
4. The number 292 in decimal system is expressed in binary system by:
(a) 10001010 (b) 100010001
(c) 100100100 (d) 10101000
[NDA 2012(1)]
5. What is the decimal number representation of the binary number $(11101.001)_2$?
(a) 30.125 (b) 29.025
(c) 29.125 (d) 28.025
[NDA 2012(1)]
6. Convert the decimal number $(57.375)_{10}$, into binary number
(a) $(111001.011)_2$ (b) $(100111.110)_2$
(c) $(110011.101)_2$ (d) $(111011.011)_2$
[NDA 2012(2)]
7. The decimal representation of the number $(1011)_2$ in binary system is:
(a) 5 (b) 7
(c) 9 (d) 11
[NDA 2012(2)]
8. The binary representation of the decimal number 45 is:
(a) 110011 (b) 101010
(c) 110110 (d) 101101
[NDA 2013(1)]
9. A number in binary system is 110001. It is equal to which one of the following numbers in decimal system?
(a) 45 (b) 46
(c) 48 (d) 49
[NDA 2013(2)]
10. The number 83 is written in the binary system as:
(a) 100110 (b) 101101
(c) 1010011 (d) 110110
[NDA 2013(2)]
11. The number 251 in decimal system is expressed in binary system by:
(a) 11110111 (b) 11111011
(c) 11111101 (d) 11111110
[NDA 2014(1)]
12. What is the sum of two numbers $(11110)_2$ and $(1010)_2$?
(a) $(101000)_2$ (b) $(110000)_2$
(c) $(100100)_2$ (d) $(101100)_2$
[NDA 2014(1)]
13. What is $(1001)_2$ equal to?
(a) $(5)_{10}$ (b) $(9)_{10}$
(c) $(17)_{10}$ (d) $(11)_{10}$
[NDA 2014(2)]
14. If $(11101011)_2$ is converted to decimal system, then the resulting number is:
(a) 235 (b) 175
(c) 160 (d) 126
[NDA 2015(1)]
15. The decimal number $(127.25)_{10}$ when converted to binary number takes the form.
(a) $(1111111.11)_2$ (b) $(1111110.01)_2$
(c) $(1110112.11)_2$ (d) $(1111111.01)_2$
[NDA 2015(1)]
16. What is $(1000000001)_2 - (0.0101)_2$ equal to?
(a) $(512.6775)_{10}$ (b) $(512.6875)_{10}$
(c) $(512.6975)_{10}$ (d) $(512.0909)_{10}$
[NDA-2015(2)]
17. What is the binary equivalent of the decimal number 0.3125?
(a) 0.0111 (b) 0.1010
(c) 0.0101 (d) 0.1101
[NDA (I) 2016]
18. If the number 235 in decimal system is converted into binary system, then what is the resulting number?
(a) $(11110011)_2$ (b) $(11101011)_2$
(c) $(11110101)_2$ (d) $(11011011)_2$
[NDA (II) 2016]
19. In the binary equation $(1p101)_2 + (10q1)_2 = (100r\ 00)_2$ where p, q and r are binary digits, what are the possible values of p,q and r respectively?
(a) 0,1,0 (b) 1,1,0
(c) 0,0,1 (d) 1,0,1
[NDA (I) 2017]
20. The remainder and the quotient of the binary division $(101110)_2 \div (110)_2$ are respectively.
(a) $(111)_2$ and $(100)_2$ (b) $(100)_2$ and $(111)_2$
(c) $(101)_2$ and $(101)_2$ (d) $(100)_2$ and $(100)_2$
[NDA (II) 2017]
21. The binary number expression of the decimal number 31 is:
(a) 1111 (b) 10111
(c) 11011 (d) 11111
[NDA (I) 2018]
22. The sum of the binary numbers $(11011)_2$ $(10110110)_2$ and $(10011x0y)_2$ is the binary number $(101101101)_2$. What are the value of x and y?
(a) $x = 1, y = 1$ (b) $x = 1, y = 0$
(c) $x = 0, y = 1$ (d) $x = 0, y = 0$
[NDA (II) 2018]
23. The binary number is represented by $(cdccddccdd)_2$, where $c > d$ what is its decimal equivalent?
(a) 1848 (b) 2048
(c) 2842 (d) 2872
[NDA (II) 2019]
24. The number $(1101101 + 1011011)_2$ can be written in decimal system as
(a) $(198)_{10}$ (b) $(199)_{10}$
(c) $(200)_{10}$ (d) $(201)_{10}$
[NDA 2020]
25. What is $(1110011)_2 \div (10111)_2$ equal to?
(a) $(101)_2$ (b) $(1001)_2$
(c) $(111)_2$ (d) $(1011)_2$
[NDA 2022 (I)]

- 26.** If $x^3 + y^3 = (100010111)_2$ and $x + y = (11111)_2$ then what is $(x - y)^2 + xy$ equal to?
 (a) $(1101)_2$ (b) $(1001)_2$
 (c) $(1011)_2$ (d) $(1111)_2$ [NDA 2022 (I)]
- 27.** What is the binary number equivalent to decimal number 1011?
 (a) 1011 (b) 111011
 (c) 11111001 (d) 1111110011 [NDA 2023 (I)]
- 28.** If $x = (1111)_2$, $y = (1001)_2$ and $z = (110)_2$, then what is $x^3 - y^3 - z^3 - 3xyz$ equal to?
 (a) $(1111001)_2$ (b) $(1001111)_2$
 (c) $(1)_2$ (d) $(0)_2$ [NDA 2025 (1)]
- 29.** What is the sum of the binary numbers $(101101101)_2$ and $(100011)_2$?
 (a) $(110010000)_2$ (b) $(110001000)_2$
 (c) $(110000100)_2$ (d) $(100100000)_2$ [NDA 2025 (2)]

ANSWER KEY

1.	b	2.	b	3.	a	4.	c	5.	c	6.	a	7.	d	8.	d	9.	d	10.	c
11.	b	12.	a	13.	b	14.	a	15.	d	16.	b	17.	c	18.	b	19.	a	20.	b
21.	d	22.	b	23.	d	24.	c	25.	a	26.	b	27.	d	28.	d	29.			

Solutions

Sol. 1. (b)

The binary number is

$$\begin{array}{ccccccc} 1 & 0 & 1 & 0 & 1 & 1 & 1 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 2^7 & 2^5 & 2^3 & 2^2 & 2^1 & 2^0 & \end{array}$$

The decimal number is $= 128 + 32 + 8 + 4 + 2 + 1 = 175$

Sol. 2. (b)

$$a = (00111)_2 = (7)_{10}$$

$$b = (01110)_2 = (14)_{10}$$

$$b/a = 14/7 = 2$$

Sol. 3. (a)

Given, decimal number $= (0.3)_{10}$

$$\Rightarrow 0.3 \times 2 = 0.6 \quad 0$$

$$\Rightarrow 0.6 \times 2 = 1.2 \quad 1$$

$$\Rightarrow 0.2 \times 2 = 0.4 \quad 0$$

$$\Rightarrow 0.4 \times 2 = 0.8 \quad 0$$

$$\Rightarrow 0.8 \times 2 = 1.6 \quad 1$$

So, its binary equivalent is $(0.01001\dots)_2$

Sol. 4. (c)

2	292	0
2	146	0
2	73	1
2	36	0
2	18	0
2	9	1
2	4	0
2	2	0
2	1	1

$$(292)_2 = (100100100)_2$$

Sol. 5. (c)

$$1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 + 0 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3}$$

$$= 16 + 8 + 4 + 0 + 1 + 0 + 0 + (1/8)$$

$$= 29 + (233/8) = (29.125)_{10}$$

Sol. 6. (a)

For integer part of 57.375 i.e., $(57)_{10}$

2	57	
---	----	--

2	28	1
2	14	0
2	7	0
2	3	1
	1	1

$$\therefore (57)_{10} = (111011)_2$$

For after decimal part of 57.375 i.e., $(0.375)_{10}$

Now,

$$0.375 \times 2 = 0.75$$

Binary

$$0$$

$$0.75 \times 2 = 1.5$$

$$1$$

$$0.5 \times 2 = 1.0$$

$$1$$

$$(0.375)_{10} = (0.011)_2$$

$$\therefore (57.375)_{10} = (111011.011)_2$$

Sol. 7. (d)

$$(1011)_2 = 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$

$$= 8 + 0 + 2 + 1 = 11$$

Sol. 8. (d)

2	45	1
2	22	0
2	11	1
2	5	1
2	2	0
	1	1

$$\therefore \text{Binary representation of } (45)_{10} = (101101)_2$$

Sol. 9. (d)

$$(110001)_2 = 1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$$

$$= 32 + 16 + 0 + 0 + 0 + 1 = (49)_{10}$$

So, the required decimal number is 49

Sol. 10. (c)

2	83	1
2	41	1
2	20	0
2	10	0
2	5	1
2	2	0
	1	1

$$\therefore (83)_{10} = (1010011)_2$$

Which is required binary form.

Sol. 11. (b)

2	251	1
2	125	1
2	62	0
2	31	1
2	15	1
2	7	1
2	3	1
	1	

$$\therefore (251)_{10} = (11111011)_2$$

Sol. 12. (a)

$$\begin{array}{r} 11110 \\ + 1010 \\ \hline \end{array}$$

$$101000$$

Sol. 13. (b)

$$(1001)_2 = (2^3 \times 1 + 2^2 \times 0 + 2^1 \times 0 + 2^0 \times 1)_{10}$$

$$= (8+1)_{10} = (9)_{10}$$

Sol. 14. (a)

$$(11101011)_2 = (?)_{10}$$

$$= [7 \times 2^7 + 1 \times 2^6 + 1 \times 2^5 + 0 \times 2^4$$

$$+ 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0]_{10}$$

$$= [128 + 64 + 32 + 0 + 8 + 0 + 2 + 1]_{10} = (235)_{10}$$

Sol. 15. (d)

$$(127.25)_{10} = (?)_2$$

2	127	1
2	63	1
2	31	1
2	15	1
2	7	1
2	3	1
2	1	1
	0	1

Here

Binary

$$0.25 \times 2 = 0.50 \quad 0$$

$$0.50 \times 2 = 1.00 \quad 1$$

$$\therefore (127.25)_{10} = (1111111.01)_2$$

Sol. 16. (b)

$$(1000000001)_2 - (0.0101)_2$$

Sol. 17. (c)

Sol. 18. (b) _____

2	2 3 5	
2	1 2 7	1
2	5 8	1
2	2 9	0
2	1 4	1
2	7	0
2	3	1
2	1	1
	0	1

Sol. 19. (a)

$$\begin{array}{r} 1\text{ p } 101 \\ + 10\text{ q } 1 \\ \hline = 100\text{ r } 00 \\ \text{p} = 0, \text{q} = 1, \text{r} = 0 \end{array}$$

$$\therefore \text{Remainder} = (4)_{10} \equiv (100)_2 \text{ and quotient} = (7)_{10} \equiv (111)_2$$

2	31	1
2	15	1
2	7	1
2	3	1
2	1	1
	0	

Sol. 22. (b)

$$\begin{array}{r} 101101101 \\ - 10110110 \\ \hline 10110111 \\ - 11011 \\ \hline 10011100 \\ \Rightarrow x = 1, y = 0 \end{array}$$

Sol. 23. (d)

$$= (1 \times 2^{11} + 1 \times 2^9 + 1 \times 2^8 + 1 \times 2^5 + 1 \times 2^4 + 1 \times 2^3) = 2872$$

$$\begin{array}{r} 1101101 \\ + 1011011 \\ \hline 11001000 = (200)_2 \end{array}$$

$$\begin{array}{r} 10111 \overline{) 1110011} \\ \underline{10111} \\ 10111 \\ \underline{10111} \\ 0 \\ \underline{10111} \\ 10111 \\ \underline{10111} \\ X \end{array}$$

$$\begin{aligned} &= x^2 + y^2 - 2xy + xy \\ &= x^2 + y^2 - xy \\ &= \frac{x^3 + y^3}{x + y} \end{aligned}$$

$$\begin{array}{r}
 11111 \overline{) 100010111} \quad 1001 \\
 \underline{11111} \\
 111 \\
 \underline{0} \\
 1111 \\
 \underline{0} \\
 1111 \\
 \underline{1111} \\
 x
 \end{array}$$

Answer = $(1001)_2$

2	1011	1
2	505	1
2	252	0
2	126	0
2	63	1
2	31	1
2	15	1
2	7	1
2	3	1
2	1	1
	0	

$$(1011)_{10} = (1111110011)_2$$

So $x^3 - y^3 - z^3 - 3xyz = 0$

$$\begin{array}{r} 101101101 \\ + \quad 100011 \\ \hline = 110010000 \end{array}$$